

1.

```
public class Node {
```

```
    Node next;  
    // Node prev;  
    int data;
```

```
    Node(int data) {  
        this.data = data;  
        this.next = null;  
        // this.prev = null;  
    }  
}
```

```
public class Stack {
```

```
    Node top;
```

```
    Stack() {  
        this.top = null;  
    }
```

```
    void push(int data) {  
        Node new_node = new Node(data);  
        if (top == null) {  
            this.top = new_node;  
            new_node.next = null;  
        } else {  
            new_node.next = this.top;  
            this.top = new_node;  
        }  
    }
```

```
}
```

```
    int pop() {  
        int temp = this.top.data;  
        this.top = this.top.next;  
        return temp;  
    }
```

```
    boolean isEmpty() {  
        if (this.top == null) {
```

```

        return true;
    } else {
        return false;
    }
}

void peek() {
    if (this.isEmpty() == false) {
        System.out.println(top.data);
    }
}

void Display() {
    Node temp = this.top;
    while (temp != null) {
        System.out.println(temp.data);
        temp = temp.next;
    }
}
}

```

```

public class Driver {
    public static void main(String[] args) {

        Stack stack = new Stack();

        stack.push(1);
        stack.push(5);
        stack.push(6);
        stack.push(6);
        stack.push(7);
        stack.push(8);
        stack.push(10);
        stack.push(38);

        stack.pop();

        stack.push(58);
        stack.push(43);

        stack.pop();
    }
}

```

```
        stack.push(100);
        stack.push(378);

        stack.Display();
    }
}
```

Q2).

1.

```
public class Node {

    Node next;
    // Node prev;
    char data;

    Node(char data) {
        this.data = data;
        this.next = null;
        // this.prev = null;
    }
}
```

```
public class Stack {

    Node top;

    Stack() {
        this.top = null;
    }

    void push(char data) {
        Node new_node = new Node(data);
        if (top == null) {
            this.top = new_node;
            new_node.next = null;
        }
    }
}
```

```

        } else {
            new_node.next = this.top;
            this.top = new_node;
        }
    }

    char pop() {
        char temp = this.top.data;
        this.top = this.top.next;
        return temp;
    }

    boolean isEmpty() {
        if (this.top == null) {
            return true;
        } else {
            return false;
        }
    }

    void peek() {
        if (this.isEmpty() == false) {
            System.out.println(top.data);
        }
    }

    void Display() {
        Node temp = this.top;
        while (temp != null) {
            System.out.println(temp.data);
            temp = temp.next;
        }
    }
}

```

```

public class Driver {
    public static void main(String[] args) {

        String st = "MADAM";
    }
}

```

```
String st2 = "ABBA";
String st3 = "abc";

System.out.println(isPalindrome(st));
System.out.println(isPalindrome(st2));
System.out.println(isPalindrome(st3));

}

static boolean isPalindrome(String in_string) {
    Stack stack1 = new Stack();

    for (int i = 0; i < in_string.length(); i++) {

        stack1.push(in_string.charAt(i));

    }
    String reversed = "";
    while (!stack1.isEmpty()) {
        reversed = reversed + stack1.pop();
    }

    if (in_string.equals(reversed)) {
        return true;
    } else {
        return false;
    }

}

}
```

2.

```
public class Node {
```

```
    Node next;  
    // Node prev;  
    String data;
```

```
    Node(String data) {  
        this.data = data;  
        this.next = null;  
        // this.prev = null;  
    }
```

```
}
```

```
public class Stack {
```

```
    Node top;
```

```
    Stack() {  
        this.top = null;  
    }
```

```
    void push(String data) {  
        Node new_node = new Node(data);  
        if (top == null) {  
            this.top = new_node;  
            new_node.next = null;  
        } else {  
            new_node.next = this.top;  
            this.top = new_node;  
        }  
    }
```

```
}
```

```
    String pop() {  
        String temp = this.top.data;  
        this.top = this.top.next;  
        return temp;  
    }
```

```
    boolean isEmpty() {  
        if (this.top == null) {  
            return true;  
        }
```

```

    } else {
        return false;
    }
}

void peek() {
    if (this.isEmpty() == false) {
        System.out.println(top.data);
    }
}

void Display() {
    Node temp = this.top;
    while (temp != null) {
        System.out.println(temp.data);
        temp = temp.next;
    }
}
}

```

```

public class Driver {
    public static void main(String[] args) {

        Stack stack = new Stack();

        String string = "ab+cd+*";
        String infix = "";

        for (int i = 0; i < string.length(); i++) {

            char c = string.charAt(i);
            if ((c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'A')) {

                stack.push(Character.toString(c));

            } else {

                char operator = c;

                String st1 = stack.pop();
                String st2 = stack.pop();

```

```
        infix = "(" + st2 + operator + st1 + " ";  
  
        stack.push(infix);  
    }  
    }  
    System.out.println(infix);  
}  
}
```